SEA Lab
Symbiotic Engineering and Analysis Lab

Carbon Sequestration Harvesting Energy from Waves:

Early Design Document

Power at Sea Prize
Department of Energy
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Short Description or Summary

As marine energy research continues to evolve, wave energy converters (WECs) have emerged as a promising technology for promoting grid resilience, powering remote and coastal communities, and supporting the Blue Economy. The SEAquestration team explores the integration of WECs with the growing demand for carbon sequestration, proposing a novel approach to marine carbon dioxide removal (mCDR). This approach, CArbon Sequestration Harvesting Energy from Waves, or CASHEW, seeks to directly utilize the mechanical energy produced by WEC oscillations to pump carbon dioxide into eligible geographies. The CASHEW system includes a WEC, pump, and pipe. Our system design assumes that CASHEW receives pre-captured CO_2, meaning that carbon capture technologies lie outside the system boundary. In this DEVELOP phase submission, we share our refined system design, updated modeling results, and location-specific case studies that demonstrate the technical and economic feasibility of CASHEW.

1. Technical Capabilities and Merit

As mentioned in the *Technical Narrative*, the DEVELOP phase of the Power at Sea Prize saw the CASHEW system concept undergo a finalization of both top-down (integral system components) and bottom-up (dependent on individual design decisions) requirements, which are summarized in **Table 1**.

The following brief analysis investigates whether the CASHEW system concept satisfies the proposed top-down and bottom up requirements. Many of the requirements necessitate the consideration of fulfillment at two separate proposed implementation locations, the Juan de Fuca Ridge and the Gulf of America, with the process of choosing these locations being covered in the *Quantitative Analysis* portion of this document.

1.1 Flow Rate Top-Down Requirement:

To determine whether the CASHEW concept can produce a flow rate at scale of 1 Gt/year of CO_2 globally [1], the utilization of the time-domain Simscape model for the system is necessary; the robustness of the model allows for many parameters to be individually assigned per-simulation—such as the wave height, wave period, sequestration depth, aquifer volume, and aquifer permeability—allowing for system performance at both proposed locations to be more accurately determined. With the appropriate parameters utilized for each simulation, the average mass flow rate of sequestered CO_2 is 0.723 Mt/year per WEC for the Juan de Fuca Ridge (off the West Coast of the US) and 0.414 Mt/year per WEC for the Gulf of America. This suggests the requirement of 1 Gt/year can be achieved with around 1383 (using the Juan de Fuca numbers) to 2415 (using the Gulf numbers) total CASHEW systems globally; these numbers could be reached with 28 - 49 sites with arrays of 50 CASHEW units each. It is also worth noting that the designs used for these results have not been optimized, and further design improvements in WEC design and PTO design could significantly reduce the required number of systems.

1.2 Time Scale Top-Down Requirement:

The time scale requirement can be satisfied by looking both at the time it takes for CO_2 to initially sequester, as well as the time period for which CO_2 will remain in the aquifer before equilibrating with the atmosphere. Due to the high capacity of basaltic rocks to trap CO_2, basalt aquifers are the most likely candidate for underwater CO_2 sequestration repositories; the high reactivity between basalt and CO_2 additionally allows for rather rapid sequestration - stabilization of CO_2 concentration in basalt occurs as soon as 96 hours, or on the order of days, after initial injection [2]. Once sequestered in an aquifer, carbon/CO_2 generally sees a time period of 300 to 1000 years before it equilibrates with the atmosphere. As there exist locations within both the Juan de Fuca Ridge and the Gulf of America regions where sequestration of CO_2 into basaltic aquifers is possible, the time scale requirement can thus be satisfied.

1.3 Pressure Top-Down Requirement:

The pressure required for sequestration is highly location-dependent. For the Juan de Fuca Ridge, gravitational trapping will be the predominant sequestration method, and requires a depth of 2700 m. For the Gulf of America, gravitational trapping remains possible if sequestration occurs around Galveston Island or in depleted oil wells, but physical trapping with caprock is also a possibility, and requires a depth of 800 m. Regardless of the method of sequestration, the pressure required in order to do so is calculated with the following equation (**Eq 1**):

Table 1: Table of requirements.

Top-Down Requirements		Bottom-Up Requirements		
Requirement:	**Value:**	**Requirement:**	**Dependencies:**	**Value:**
Flow Rate	1 Gt/yr of CO_2 at scale	**Pumping System Energy**	Method, Phase, and container	≥ power required for sequestration
Time Scale	To sequestration: ~ days Stability maintained for: > 100 years	**Temperature, Pressure, Density, Depth**	Phase: Supercritical CO_2	31 ℃, 7.38 MPa, 850 kg/m^3, > 800 m
Pressure	Sufficient for desired sequestration method	**Depth**	Method of Sequestration: Gravitational	> 2700 m
Pipe Performance	Corrosion manageable during system lifetime Pipe does not fail due to hydrostatic pressure at depth	**Geothermal Gradient/Salinity Relationship**	Method of Sequestration: Gravitational	Obeys **Eq 5.**
Lifetime Cost	$6-12 per ton of CO_2	**Grain Size Area**	Rock container: Basalt	0.81 mm^2 (Plagioclase)
Location Wave (Power) Availability	≥ power required for sequestration	**Dissolution Rate**	Rock container: Basalt	High dissolution rate
Environmental Disruption	CO_2 must not disrupt deep sea ecosystems	**Temperature**	Rock container: Basalt	High temperature, and dependent on specific type of basalt
-	-	**Pressure**	Rock container: Basalt	101.325 kPa (1 atm) to a few hundred MPa

$$P_{floor} = P_{ocean} + P_{frack} = \rho_w g z_{floor} + 700(z_{injection} - z_{floor})^{1.5} \qquad \textbf{Eq 1.}$$

where P_{floor} is the total pressure required, P_{ocean} is the hydrostatic pressure at the desired depth, and P_{frack} is the pressure required to fracture the seabed for sequestration. The pressure calculations utilize the density of seawater ($\rho_w = 1025\ kg/m^3$), the acceleration due to gravity (g), the ocean depth (z_{floor}) and the injection depth ($z_{injection}$, assumed to be 200 m greater than the ocean depth). The pressure required for gravitational and physical trapping sequestration is 29.129 MPa and 10.024 MPa, respectively. The piston area can be designed to match wave excitation force to required pressure. But in higher excitation force sea states, larger pistons can be used which results in higher sequestration rates.

1.4 Pipe Performance Top-Down Requirement:

Given that the proper steel alloy is used for the pipe system of the CASHEW concept, the lifetime of the pipe should far exceed that of the anticipated lifetime of CASHEW: 316 stainless steel is our recommendation. Additionally, the pipe will experience a pressure differential at depth of a maximum of 1.98 MPa (seen at the Juan de Fuca Ridge); utilizing the following equivalent stress (σ_{equiv}) and radial pipe deformation (δ) equations with 316 stainless steel material properties, a maximum equivalent stress of 27.72 MPa and deformation of 0.02 mm will result, which are both small enough to conclude that the pipe will survive at depth:

$$\sigma_{equiv} = \frac{\pi D}{2t} \qquad \textbf{Eq 2a.}$$

$$\delta = \frac{D}{2E} \times \frac{PD}{2t - \frac{\nu PD}{4t}}$$

Eq 2b.

with pipe inner diameter D = 0.26 m, pipe thickness t = 0.01 m, pressure P = 1.98 MPa (the differential), Young's Modulus E = 195 GPa, and Poisson's ratio ν = 0.265.

1.5 Lifetime Cost Top-Down Requirement:

The lifetime cost per ton of CO_2 will be denoted as the LCOC, or Levelized Cost of Carbon, for the CASHEW concept. It is calculated via the following equation:

$$LCOC = \frac{Annual\ Lifetime\ Cost}{Annual\ Carbon\ Sequestration}$$ **Eq 3.**

A more thorough LCOC calculator is shown in the Quantitative Analysis, section 2.3. The LCOC calculated for the two locations are \$6.30/tCO_2 for the Juan de Fuca ridge, and \$10.21/tCO_2 for the Gulf of America location. These LCOC values both meet the required \$6-12 per ton of CO_2 at scale.

1.6 Location Wave (Power) Availability Top-Down Requirement:

To determine if the available wave power is enough for successful CO_2 sequestration at a meaningful rate, our system dynamics model (described in detail in the Quantitative Analysis, section 2.2) is used. It is worth noting that by sizing the WEC-driven piston, we are able to set the effective "gear ratio" of the system. This means that with smaller WEC pistons, we see a greater mechanical advantage, with larger pressures reached with smaller forces. Therefore, regardless of pressure requirements, a piston can be sized sufficiently small that will enable sequestration. However, with smaller pistons, smaller sequestration rates are reachable. With a global sequestration goal of 1 Gt/year, we do not expect any one CASHEW system to achieve that, rather, a global fleet should reach this goal. Since we currently only evaluate 2 possible locations, we look for sequestration rates that would require a global deployment in the ~1000s of units or less (to be clear, we do not believe ~1000s of sites is realistic, or one array of ~1000s of WECs, but instead recommend ~20s of sites with ~50 WECs each). If hundreds of thousands of systems are required, that would indicate a system that sequesters too little to be considered a large-scale global solution. However, such a system could still have benefit in other localized systems if still found economically advantageous. This requirement couples the two requirements of pressure and flow though, where pressure effectively creates a size maximum on the piston, and flow rate creates a minimum, while this requirement asks, is the combination of the two requirements possible?

1.7 Environmental Disruption Top-Down Requirement:

Given the limited leakage potential of CO_2 sequestered in basaltic aquifers, when CO_2 is sequestered, there will be limited disruption to the environment surrounding the aquifer. As CO_2 may remain stable in aquifers for between 300 to 1000 years, environmental disruption may not occur before that amount of time has passed [3]. As CO_2 will not be directly injected into the water column, there should be limited effects to the surrounding environment.

1.8 Phase-Dependent Bottom-Up Requirements:

Per [4], an energy of 90 kWh/t and 93 kWh/t was necessary to sequester liquid and supercritical CO_2, respectively. Given the power requirements and flow rate at each location, the energy used to sequester 1 ton of CO_2 can be determined via the following equation:

$$Energy = (Power)/(Mass\ Flow\ Rate)$$ **Eq. 4**

This yields an available energy per ton CO_2 of 10.71 kWh/t and 4.94 kWh/t for the Juan de Fuca Ridge and Gulf of America, respectively. The CASHEW system is able to sequester CO_2 at a more efficient rate than electric systems presented in the studies.

To maintain its supercritical nature, supercritical CO_2 must remain at a temperature of 31 ℃, a pressure of 7.38 MPa, a density of 850 kg/m^3, and must be sequestered below 800 m. These requirements are inherently met with the underlying assumptions of the model, as the temperature remains constant, and the pressure and density increase as depth increases (and are always greater than or equal to the necessary values); sequestration also occurs at depths below 800 m for both the Juan de Fuca Ridge and the Gulf of America.

1.9 Method-Dependent Bottom-Up Requirements:

Gravitational sequestration is only possible at depths greater than 2700 m. Sequestration at the Juan de Fuca Ridge aquifer will be of the gravitational method, which is possible as the aquifer is located at a depth of 2864 m. Sequestration at the Gulf of America location may not be gravitational, as its depth remains less than 2700 m (1833 m).

The locations at both the Juan de Fuca Ridge and the Gulf of America were chosen to additionally adhere to the following relationship between the density of water and CO_2:

$$\begin{aligned}\rho_{CO2} - \rho_{H2O} = & -6.9542 - 2.8308(T - T_0) + 2.5402(P - P_0) \\ & - 0.75503(S - S_0) - 0.031177(P - P_0)^2 \\ & + 0.038104(T - T_0)(P - P_0) \\ & + 0.001448(T - T_0)(S - S_0) \\ & + 0.00046729(P - P_0)^3 \\ & + 0.00020083(T - T_0)^2(P - P_0) \\ & - 0.00070787(P - P_0)^2(T - T_0)\end{aligned} \qquad \textbf{Eq 5.}$$

where T_0 = 288 K, P_0 = 40 MPa, S_0 = 35 psu, valid over the range T = 273-301 K, P = 27-55 MPa, and S = 32-60 psu [5].

1.10 Rock Container Bottom-Up Requirements:

Given basalt as the aquifer medium, high reactivity with CO_2 is a necessity for short sequestration times. For this to occur, there should be an abundance of the mineral plagioclase in any given aquifer, as it has the highest dissolution rate for carbonation relative to other basaltic minerals. Additionally, the grain size area of the mineral should be small, and per Rasool and Ahmed [6] a sufficient surface area would be less than 0.81 mm^2 for plagioclase, which typically has higher grain sizes than other basaltic minerals. While higher temperatures of basalt greater facilitate reactions between Plagioclase and CO_2, this may be difficult to manage when the basalt aquifer is located kilometers underwater. However, if possible, the temperature at the aquifer should be as high as can be achieved. Depending on the specific composition of the basalt aquifer, higher pressures may favor higher reactivity with CO_2, thus an aquifer pressure of 101.325 kPa to a few hundred MPa must be maintained; this should be quite possible given the hydrostatic pressure at the aquifers considered at both the Juan de Fuca Ridge and the Gulf of America.

2. Quantitative Analysis

While CASHEW's technical requirements have largely been determined, a variety of factors such as sequestration method, deployment location (with respect to both wave power and geological conditions), piston model and dynamics, pipe dynamics, etc., need to be evaluated to assess our design's sequestration potential. Since CASHEW uses direct drive, it relies upon sufficient wave power available at the deployment site to pump captured CO_2 into identified saline aquifers. With in mind, the primary governing equation can be described as follows:

$$Power = Q_{CO2}\ P_{injection} \qquad \textbf{Eq 6.}$$

where Q_{CO2} represents the volumetric flow rate of CO_2, $P_{injection}$ represents injection pressure of CO_2, and *Power* represents the power required to pump CO_2. **Eq 6.** will determine the necessary power value to pump CO_2 into identified regions.

Before identifying eligible regions for deployment, our group first sought to establish the methods by which carbon will be achieved. As discussed in *Technical Narrative*, our group has pivoted from fracking and will now pursue aquifers as the sequestration medium with physical and/or gravitational trapping as the method.

1. *Physical Trapping (Structural Trapping):* CO_2 is physically trapped by a cap rock, which seals CO_2 by virtue of having low permeability. Aquifer depth must be 800 meter below the sea bed for the CO_2 to be maintained or transitioned in the supercritical state.
2. *Gravitational Trapping:* Uses density difference between CO_2 and seawater at a depth of at least 2700 meters.

With these two potential sequestration methods established, as well as a method of modeling the required wave availability (**Eq 6.**), we can determine eligible locations and relevant CASHEW outputs.

Table 2: CO_2 trapping options with associated depth and phase requirements.

	Location	Depth (from waterline)	Phase	Co-location with basalt?
Aquifer + Structural	Aquifer with cap rock	800m	Supercritical	Possible
		N/A	Cold liquid	
Aquifer +	Aquifer with viable geothermal	>2700m	Cold liquid	Possible

Gravitational	gradient and salinity			

2.1 Eligible Locations

To determine where CASHEW can be deployed, available wave power and location of possible saline aquifers needs to be established. These factors, once known, can help determine where there is both adequate wave power and proper geologic locations, thereby enabling CASHEW power production. Once a site has been selected, the theoretical mass flow rate of CO_2 that can be successfully pumped and sequestered can be ascertained.

To evaluate potential deployment sites, Geographic Information System (GIS) analysis using QGIS was utilized to map wave power availability in identified coastal regions (Juan de Fuca and the Gulf of America). Within these broad regions, we then mapped the location of various saline aquifers, noting their permeability, basalt availability, and depth requirements. Using these two mapping methods, we were then able to identify a specific location (longitude and latitude) with adequate wave power and aquifer characteristics to enable either physical or gravitational trapping.

2.1.1 Wave Power Availability:

To determine possible regions for CASHEW's operation, we first conducted research to determine regions in/around the US that contain saline aquifers. While saline aquifers have a significant presence within the US, much of existing aquifer literature focuses on freshwater aquifers for groundwater applications. These aquifers are typically located inland or near lakes– neither of which can be integrated with wave power. This limitation initially restricted our search for suitable locations, prompting us to explore Juan de Fuca and the Gulf of America.

Please note that throughout this analysis, significant wave height refers to the average wave height from crest to trough of the highest one-third of the waves in a specific time period, peak wave period refers to the wave period associated with most energetic waves in the spectrum for a specific time period, and wave power refers to energy per unit length generated by the movement of ocean waves. Wave power is calculated using **Eq 7.**

$$P = \frac{\rho g^2}{64\pi} * H^2 * T \qquad \textbf{Eq.7}$$

where ρ represents the density of water (kg/m^3), g represents the acceleration due to gravity (m/s^2), H represents the significant wave height (m), and T represents the peak period (s).

<u>Location 1: Juan de Fuca:</u>

The Juan de Fuca region refers to a tectonic plate off the coast of Oregon and Washington. It is located near a seismic ridge, meaning that the region is basalt-rich. Geographic and aquatic conditions in this region are suited for both physical (structural) and gravitational trapping, as surrounding aquifers meet depth and salinity requirements (>2700m in depth, and salinity below 100,000–200,000 ppm).

To visualize the wave power available in the Juan de Fuca region, we utilized NREL's Marine Energy Atlas [7]. This interactive GIS platform allows us to visualize annual wave power availability in kW/m across the identified region **(see Figure 1).** Based on NREL's data visualization, annual wave power ranges between 4 kW/m near shore to 56 kW/m when 300 km offshore. In the identified area, wave power is an average of 40-48 kW/m.

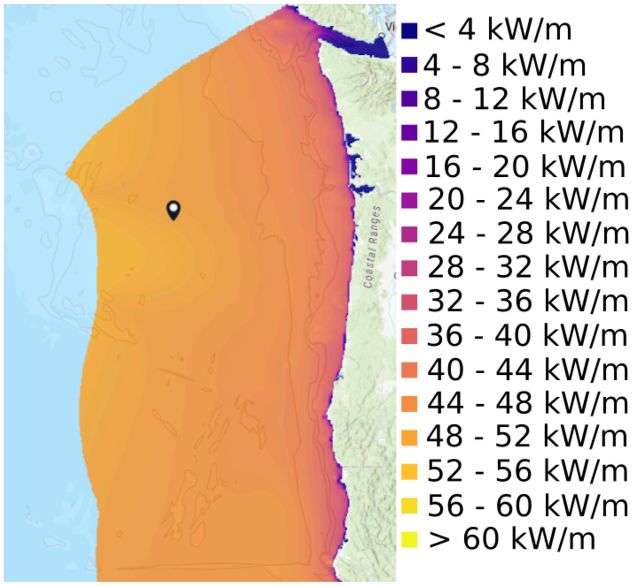

Figure 1: Juan de Fuca overall wave power distribution [7].

<u>Location 2: Gulf of America</u>

The Gulf of America region refers to an oceanic basin/marginal sea of the Atlantic Ocean. For this analysis, we considered oceanic regions up to 300 km away from the shoreline. The Gulf, unlike Juan de Fuca, is not a region of high seismic activity. Gulf waters are characterized by significant levels of salinity. In fact, its seafloor is dominated by large salt-structures, which over time have migrated upward, yielding features like salt domes that influence the seafloor's topography and sediment deposition. Given the Gulf's high levels of salinity, this region is not suited for gravitational trapping, as CO_2 is less soluble in such waters.

To visualize the wave power available in the Gulf of America region, we could not employ the same NREL tool, as NREL does not provide datasets in this region, as shown below. Additionally, NREL's publicly-available GitHub repository also does not provide wave data for the Gulf Region.

Given this constraint, additional sources such as the United States Geological Survey (USGS) were utilized. In the USGS publication "Climatological Wave Height, Wave Period and Wave Power along Coastal Areas of the East Coast of the US and Gulf of Mexico," [8] spatial variations in climatological wave parameters (ie significant wave height, peak wave period, and wave power) are compiled for areas along the US East Coast and Gulf of America. Using this data, a QGIS visualization was developed **(see Figure 2).** In this figure, annual wave power ranges from 0 W/m onshore to 2,615 W/m approximately 250 km offshore. While wave power availability varies along the Gulf coast, a "blue sliver" becomes apparent, indicating a region of high wave power availability. Within this sliver, power values range from 1.298-2.615 kW/m. Upon further inspection, this sliver corresponds to regions up to 250 km off the coast of Texas, specifically along San Jose, Mustang, and Padre Islands (see "Identified Region" in **Figure 2**).

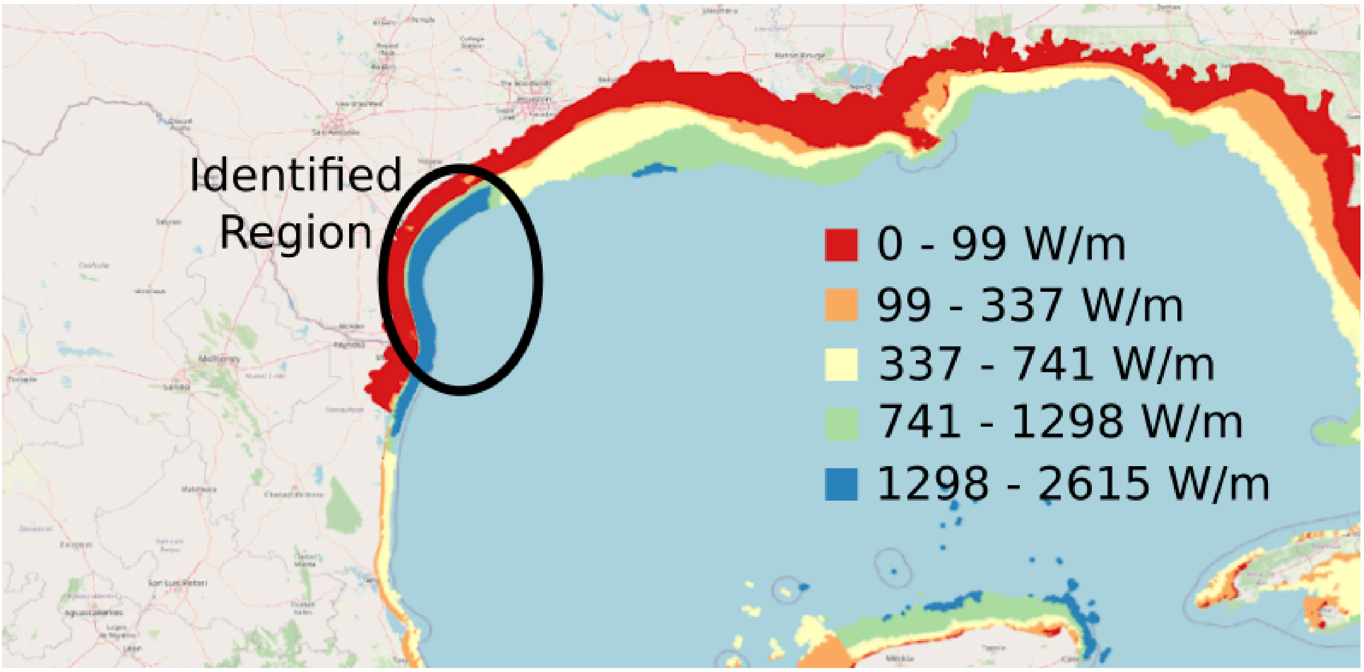

Figure 2: Gulf of America overall wave power distribution

Through this analysis, we have identified regions within the Juan de Fuca and Gulf of America regions with significant wave power. It is important to note that while both regions have sufficient wave power for WEC operation, the average wave power in the Juan de Fuca region is roughly twenty times that of the Gulf of America (40 kW/m vs 2.615 kW/m). This indicates that Juan de Fuca may be an advantageous location for multiple WEC-based applications.

2.1.2 Aquifer Locations:

Location 1: Juan de Fuca

In the previous step, we have identified general wave power availability within the Juan de Fuca region. With this in mind, we now need to identify potential saline aquifers. According to the study "Carbon Dioxide Sequestration in Deep-Sea Basalt," [9] the Juan de Fuca region includes an aquifer area of approximately 78,000 km^2 that meets the depth requirement of 2,700 meters and a sediment cover of 200 meters. These criteria make the region suitable for both physical and gravitational trapping mechanisms.

To visualize eligible aquifers, the team once again utilized QGIS **(Figure 3).** Using NREL Marine Energy Atlas datasets, bathymetry and plotted contour lines information could be incorporated into the visualization. Knowing our minimum depth requirement was 2,700m, we plotted all viable depths in green and unviable depths (those that were too shallow) in red. Studies on carbon sequestration in deep-sea basalt by Goldberg et al have found the shaded region to be an area with basalt available for sequestering carbon [9]. In **Figure 3,** the two circular dots (1 green and 1 red) represent two regions in which there is permeability data from "Permeability within basaltic oceanic crust." [10] Given the upper dot is in an area of insufficient depth, it is denoted with a red marking, and is thereby eliminated. This leaves the green dot, as it has permeability data, meets the depth requirement of being over 2,700 meters, and overlaps with basalt. Given this met criteria, the green dot, located at (45.75, -127.76), was determined to be our ideal location for the Juan de Fuca Aquifer. This identified region is therefore used for further analysis of our system.

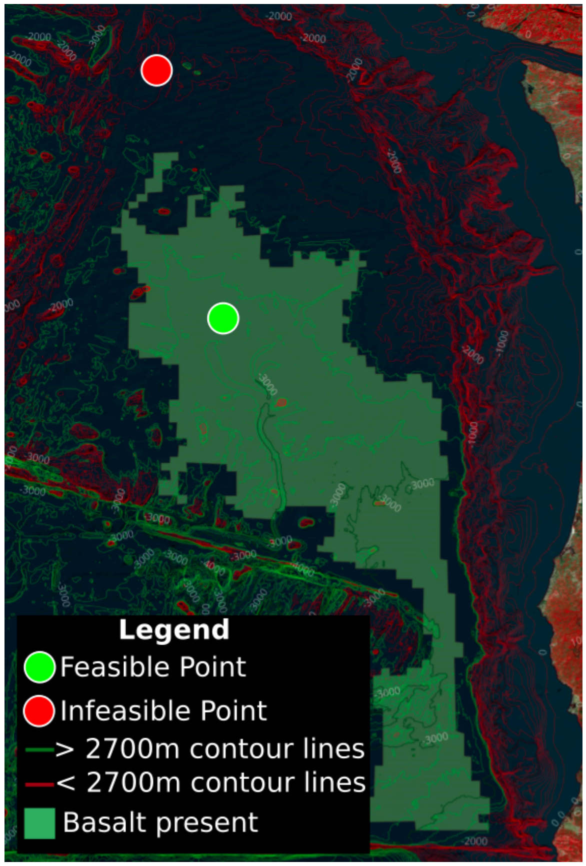

Figure 3: Map of the Juan de Fuca ridge with permeability locations (circles), basalt availability (green shading), and depth requirements (lines) highlighted.

Location 2: Gulf of America

Saline aquifers are found throughout the Gulf of America region, particularly in the Mississippi embayment and the coastal lowlands **(see Figure 4).** Despite this system of aquifers, existing literature primarily focuses on groundwater aquifers such as the Gulf Coast Aquifer, which extends along the coastline from Louisiana to Mexico. This system is composed of individual aquifers, including the Chicot, Evangeline, and Jasper. Despite the literature gap, the Bureau of Ocean Energy Management (BOEM), as well as NREL have begun looking into reservoir-based sequestration. BOEM primarily provides geologic data, but the organization has been prioritizing investigation of depleted oil wells for CO_2 sequestration; given this priority, the organization does not directly track the reservoir type when listing data, but rather lists the “sand” or “play.” In this context, “sand” refers to an offshore sand resource area managed by BOEM within the Marine Minerals Program. Sands listed within a region correspond to Atlantic OCS Aliquots with Sand Resources [11] or Marine Minerals Information System [12]. “Play,” on the other hand, refers to a geological area with potential for oil and gas accumulation. While BOEM does not explicitly track whether an identified region is an aquifer, studies such as Wendt et al (2022) “A multi-criteria CCUS screening evaluation of the Gulf of Mexico, USA,” have overlaid datasets from both BOEM and NREL to identify optimal regions for saline-based CCUS storage [13].

Wendt et al. (2022) develops a screening method for ranking viable offshore CO_2 storage sites, running four scenarios with different priorities— Scenario 1 (S1): Long-term storage in the most favorable geologic settings, Scenario 2 (S2): CO_2 storage through EOR with emphasis on maximum oil return, Scenario 3 (S3): Long-term storage prioritizing reuse of existing infrastructure for cost savings/reducing logistical burden, and Scenario 4 (S4): CO_2 storage through EOR with emphasis on reuse of existing infrastructure for cost savings/reducing logistical burden. For our analysis, Scenario 1 was best suited for our goals.

To identify optimal storage locations in Scenario 1, authors developed an offshore geologic CO_2 storage database used in conjunction with an existing Office of Fossil Energy (FE)/NETL Offshore CO_2 Saline Storage Cost Model. In their model, offshore stratigraphy was aggregated from publicly-available datasets (primarily the NETL GOM Reservoir Sands Database [14] and geophysical well logs) into stratigraphic units using stratigraphic nomenclature (sands, plays) common to the BOEM Atlas of Gulf of Mexico Gas and Oil Sands Data [15]. Using play data, the authors mapped potential saline storage reservoirs and ranked them based on thickness, porosity, permeability, depth, temperature, pressure, etc. From this ranking, authors developed **Figure 5,** where dark blue regions represent high suitability for geologic storage. Authors also published **Figure 6**, indicating that saline reservoirs are present through the regions explored in **Figure 5.** This is corroborated by **Figure 7**, which indicates the presence of the central lowlands aquifer system.

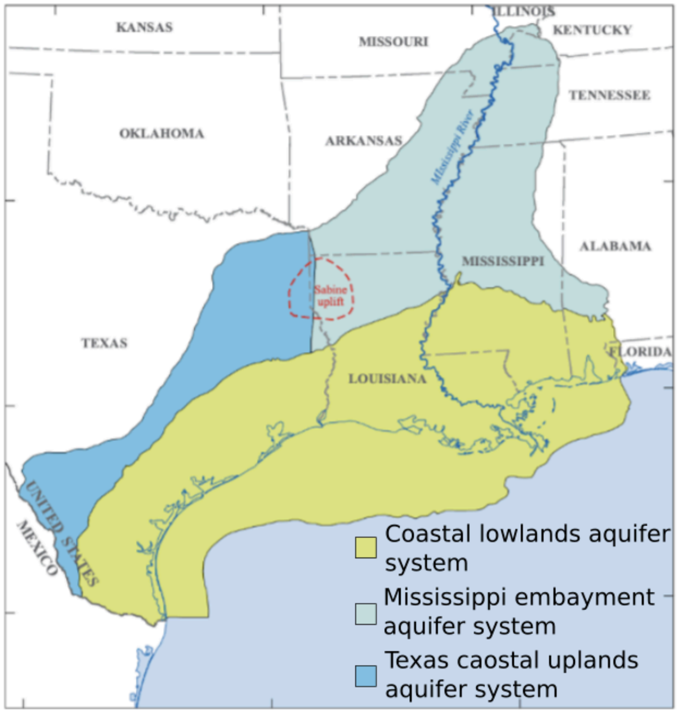

Figure 4: Saline aquifer presence along Gulf Coast [16]

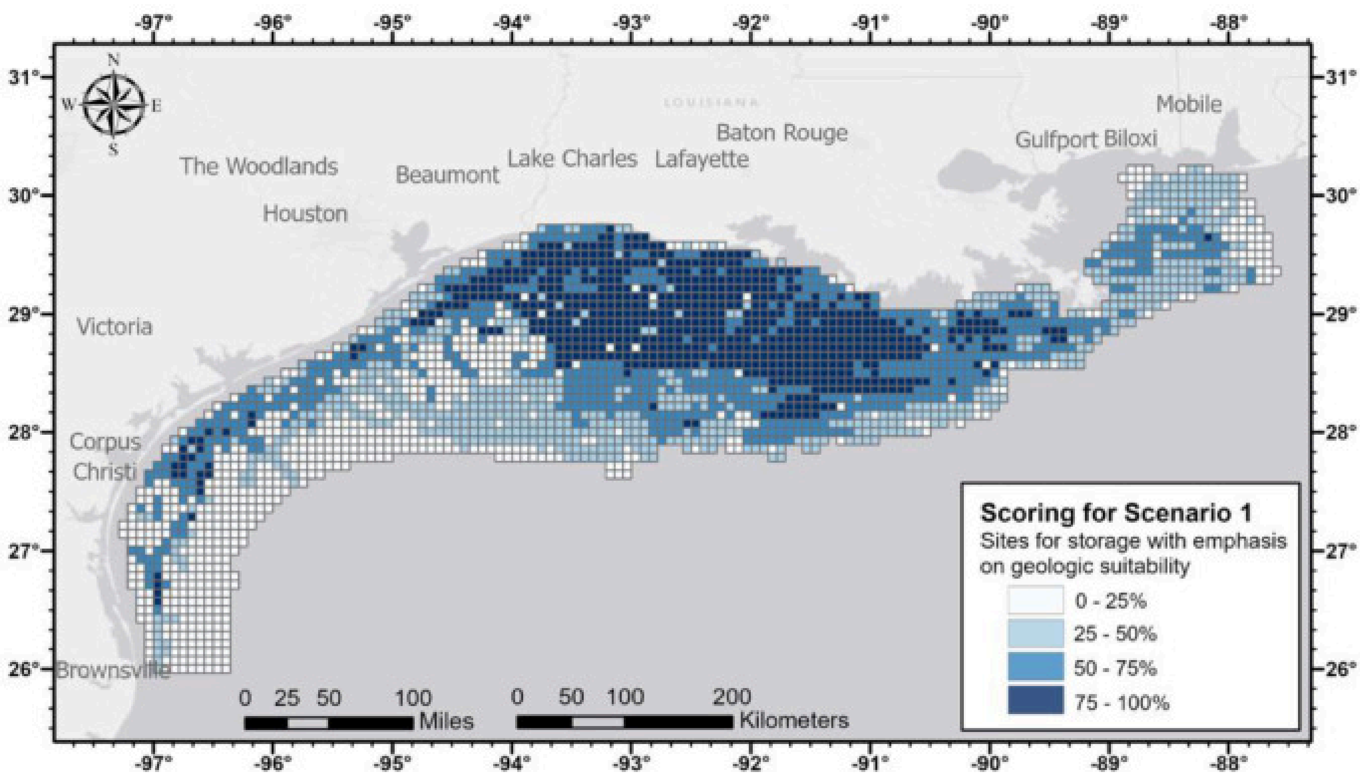

Figure 5: Scenario 1 Results [13]

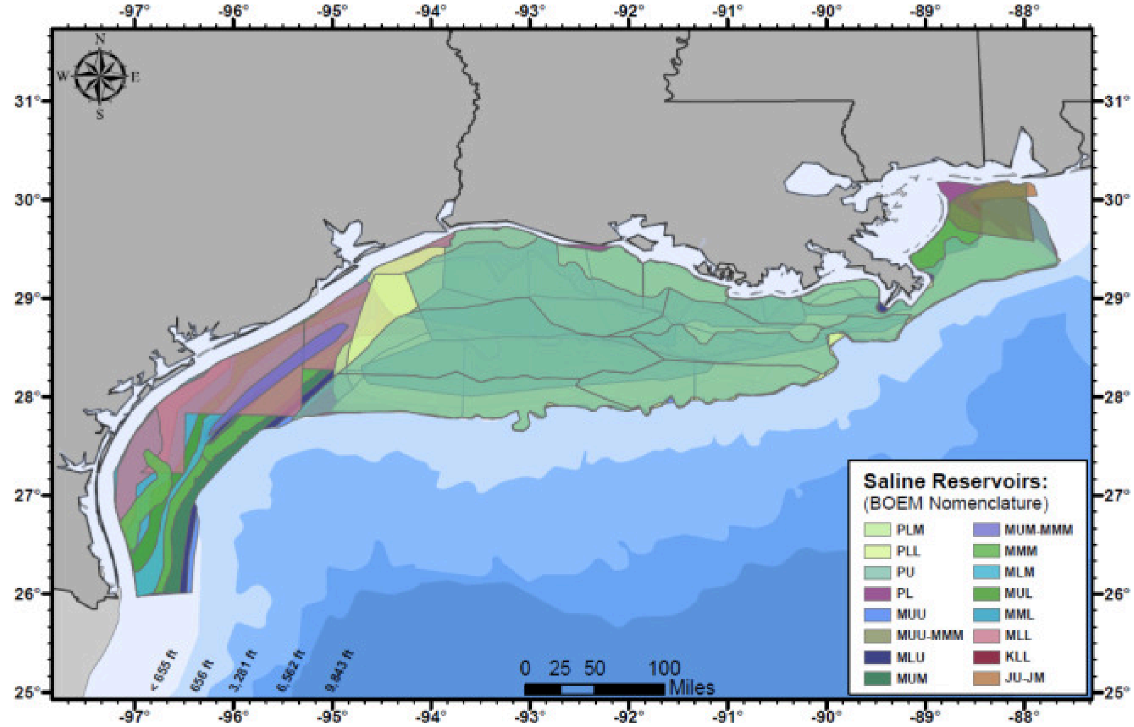

Figure 6: Ensuring presence of saline aquifers in identified region [13]

Based on wave power analysis conducted in the previous section (2.1.1), we saw the highest densities of wave power in the "blue sliver," or the region offshore Jose, Mustang, and Padre Islands. Within this region, we see in **Figure 5** there are a few regions of high storage capacity (i.e. off the coast, near Corpus Christi). While **Figure 5** indicates a high concentration of storage potential in the Louisiana Basin, our wave power analysis revealed a lower wave power availability in this region. Given CASHEW's emphasis on direct drive, we ultimately decided to investigate regions with high wave power density, leaving the cluster of blue off the shore of Corpus Christi in **Figure 5** This region corresponds to coordinates ~26.5 in latitude and ~ -96 in longitude.

While Wendt et al did not publish their developed model, our group used the same BOEM database [15] used to identify aquifer characteristics. Of the eligible coordinates in the region identified, we chose (26.67805, -96.98727), corresponding to MMM (Middle Middle Miocene) in BOEM's database. This was further validated with **Figure 6.**

2.1.3 Location Takeaways

Based on wave power and aquifer availability explored above, we selected a specific location of interest in both the Juan de Fuca and Gulf of America regions. Using these locations, the following wave power and aquifer attributes can be elucidated. For both Juan de Fuca and Gulf of America, wave data comes from QGIS analysis conducted in section 2.1.1, Wave Power Availability. For Juan de Fuca, aquifer characteristics are determined from sources listed in section 2.1.2, Aquifer Locations. For Gulf of America, aquifer data is sourced from BOEM's database at (26.67805, -96.98727); this corresponds to Sand Sequence Number of 415646, or a Sand Name of 2161_PN1010_7150. The key highlights of the location study are shown in **Table 3**.

For the Juan de Fuca site location, mCDR competitors such as Ebb Carbon are also interested in the Sequim Bay area, which is very close to Juan de Fuca. Ebb Carbon is based in this location, as the initiative is based at the Pacific Northwest National Laboratory (PNNL) in Sequim. This project is a collaboration between PNNL, University of Washington, and NOAA. Working in the same location as Ebb Carbon may be useful, as our approaches to carbon sequestration are vastly different– we could learn from each others' methods.

In the Gulf of America, there is considerably less private-venture mCDR activity. Despite this, the University of Texas at Austin is investigating use of acoustic methods for marine carbon dioxide removal via blue carbon burial in seagrass meadows. Like Ebb Carbon, working in the same location as another mCDR company may be advantageous, as we have different approaches to mCDR and could therefore learn from each other.

Table 3: Wave power and aquifer attributes for the case study locations.

	Juan de Fuca: (45.75, -127.76)	Gulf of America: (26.67805, -96.98727)
Average Peak Period	12.11 s	6.431681 s
Average Significant Wave Height	3.14 m	1.607645 m
Average Wave Power	61.8 kW/m	1.6507 kW/m
Permeability	$5 - 9\, x\, 10^{-12}\ m^2$	$2.17\ x\, 10^{-13}\ m^2$
Depth to Seabed	2864 m	1832 m
Aquifer Thickness	1 m	17.8 m
Aquifer Volume	$7,800\ km^3$	$17.3\ x\, 10^{-3}\ km^3$
Aquifer Pore Volume	$780\ km^3$	$4.8\ x\, 10^{-3}\ km^3$
Carbon Storage Potential	208 Gt liquified carbon	3 Mt liquified carbon

2.2 System Dynamics Model

To determine the yearly sequestration rate of an individual CASHEW system, a system dynamics model that determines the flow rate of CO_2 into the aquifer based on environmental parameters and system design is necessary. To accurately model this system, coupled dynamics between the WEC, PTO, and aquifer are necessary to fully characterize system performance. To model the coupled dynamics of the system, the model we presented in the CONCEPT phase is integrated with a WEC model leveraging WEC-Sim [17]. WEC-Sim enables time domain modeling of the hydrodynamics, shown in the equation below:

$$\xi \boldsymbol{M} = \boldsymbol{f_e} + \boldsymbol{f_r} + \boldsymbol{f_b} + \boldsymbol{f_u}(\xi, t) \qquad \textbf{Eq 8.}$$

where ξ represents the WEC motion, $\boldsymbol{M}$ represents the mass/inertia matrix of the WEC, $\boldsymbol{f_e}$ represents the wave excitation force, represents the radiation force (due to added mass and radiation damping), represents the hydrostatic forces (due to buoyancy and gravity), and $\boldsymbol{f_u}(\xi, t)$ represents the force on the WEC due to the PTO, in this case the direct drive sequestration system. Because of WEC-Sim's implementation in Simulink and Simscape, we can easily pass WEC motion into the existing PTO Simscape model from the CONCEPT phase, shown in **Figure 7**.

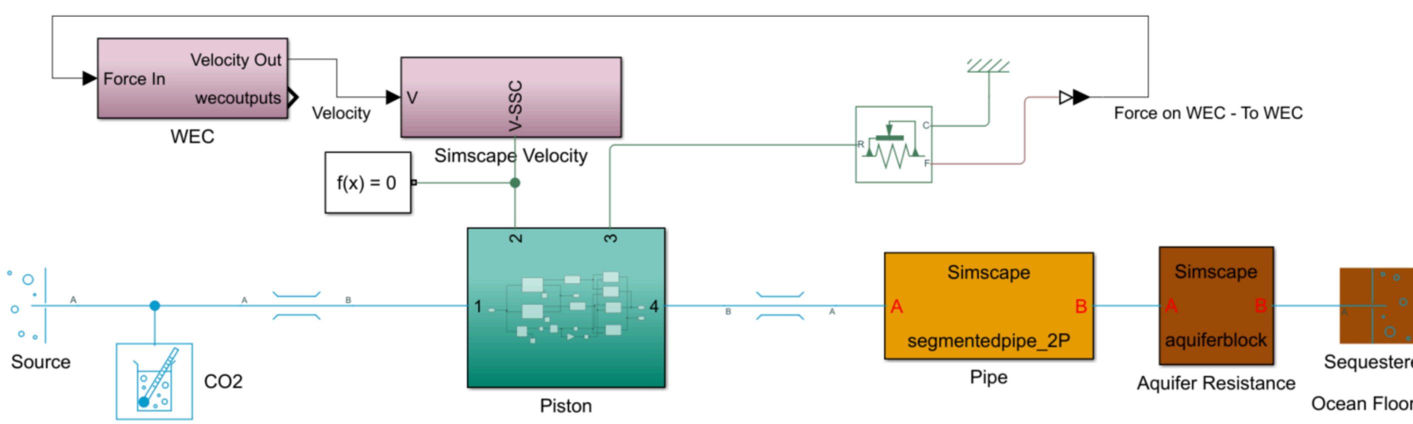

Figure 7: shows the coupled WEC-PTO model. $\boldsymbol{f_u}(\xi, t)$ is calculated given the velocity signal, and the calculated force is returned to the WEC model.

The WEC architecture used in this study is a heaving point absorber, the RM3 reference model. The Simscape implementation and a WEC-Sim generated figure of the WEC are shown in **Figure 8**. Because the WEC-Sim model outputs a Simulink velocity signal rather than a Simscape velocity, the "Simscape Velocity" block handles the proper conversion.

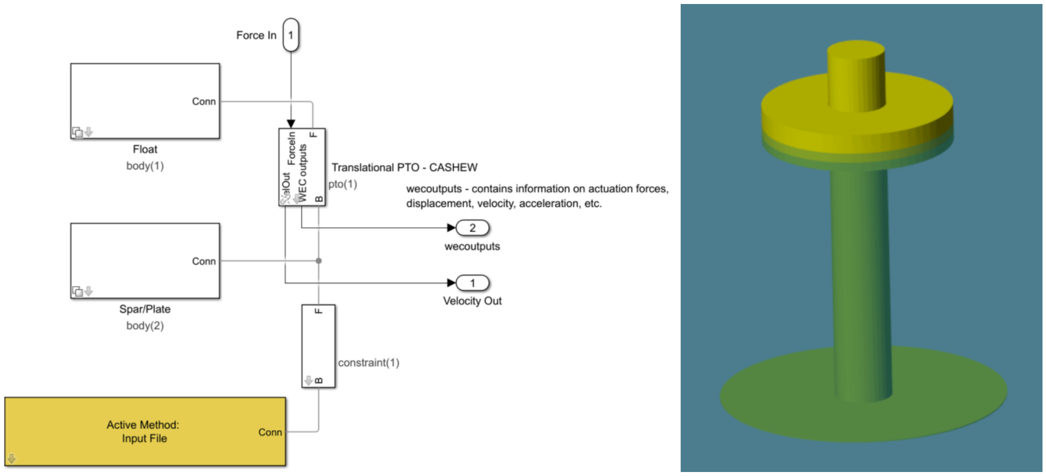

Figure 8: shows the WEC Simscape model with constraints as well as the WEC-Sim image.

The Simscape velocity signal then feeds into the “Piston” block, shown in **Figure 9**. The piston model is unchanged from the CONCEPT phase and still features the two pistons that together represent a single double acting piston, along with the directional valves that ensure the piston intakes from the carbon source, and outputs on the sequestration side.

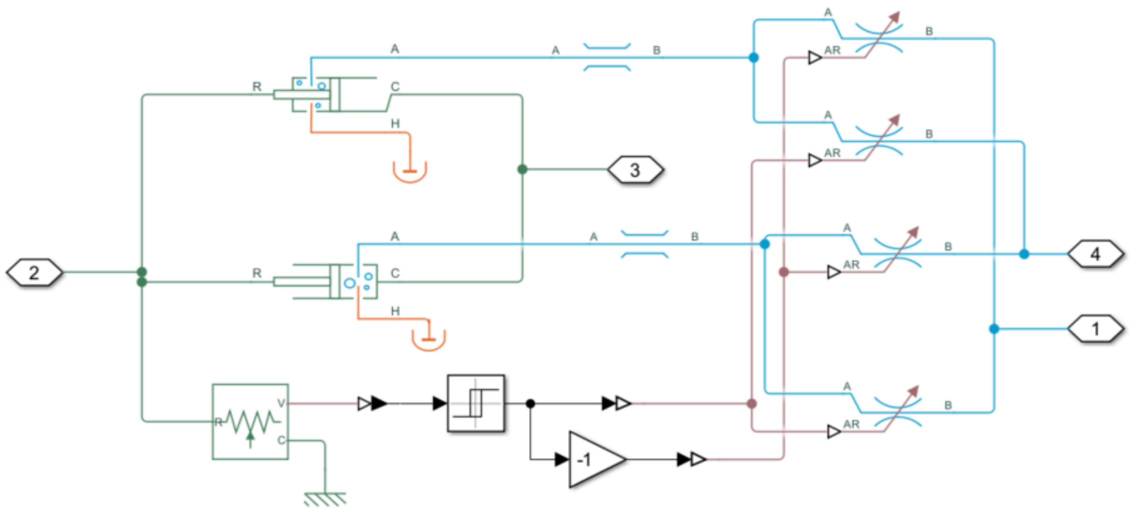

Figure 9: Simscape model of the piston and directional valves.

Flow out of the piston on the sequestration side is then modeled flowing through the long vertical pipe block labeled “Pipe”. This model is also unchanged from the CONCEPT phase, and features many segments of vertical pipes, which model frictional losses along with gravity impacts. Like in the CONCEPT phase, a convergence study was performed to identify the minimum number of segments required (40) to accurately produce the results. Flow out of the pipe is then sent to the aquifer model, which marks a major deviation from the work presented in the CONCEPT phase, where we modeled fracking instead of aquifer permeability. Like with the long vertical pipe component, the Simscape Fluids toolbox does not come with a prepackaged model of an aquifer, so we built a model based on Darcy’s law as applied to aquifers:

$$\Delta P = \frac{Q\mu}{Kh} \qquad \textbf{Eq 9.}$$

where ΔP represents the change in pressure (MPa), Q represents the volumetric flow rate (m^3/s), μ represents the viscosity (Pa-s), K represents the permeability (m^2), and h is the aquifer thickness (m). The change in pressure is then calculated as the difference between the pressure at the bottom of the pipe and the pressure in the aquifer, represented as a static pressure in this model. It is worth noting that although in this current iteration of the model, aquifer pressure is modeled as constant, the pressure would increase the lifetime of the system. However, this change is negligible on the time scale the system dynamics simulation models (~ minutes).

Aquifer parameters like permeability (K) and sea state parameters like significant wave power and peak period are provided to the model from the location study described in section 2.1.

2.3 Economics

To evaluate the economic viability of the CASHEW system, a levelized cost of carbon (LCOC) is computed using the following equation:

$$\text{LCOC} = \frac{(\text{CapEx} \times \text{Capital Recovery Factor}) + \text{Annual OpEx}}{\text{Annual Carbon Sequestration (tons CO}_2)} \quad \textbf{Eq 10.}$$

where "Annual Carbon Sequestration" is calculated using the system dynamics model shown earlier. "CapEx" and "Annual OpEx" represent the upfront cost and annual operation and maintenance costs respectively. The "Capital Recovery Factor" is calculated using the following equation:

$$\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} \quad \textbf{Eq 11.}$$

where n is the system lifetime (20 years) and r is the discount rate, which was estimated to be 6.5% based on an NREL paper [18] that used a conservative estimate of 7% and an optimistic estimate of 6%.

For the CapEx and OpEx, these were split into 3 groups: WEC, piston, and pipeline. For the WEC, the cost was determined using the RM3 technical report [19]. It is important to note that many costs could be ignored here, such as PTO costs, since our system does not need to generate electricity. For the piston costs, the cost of the piston included in the RM3 technical report was used. A piston cost model that accurately reflects piston design decisions is set for future work since the design requirements for this piston are quite different (smaller, but with a more corrosive working fluid).

For the pipeline (the large pipe that goes from the WEC at the surface to the seafloor) the cost is based on a report on offshore pipeline construction costs [20], then adjusted for inflation and the length of the CASHEW pipeline.

3. Systems Analysis

For the Juan de Fuca deployment site there is an average of 61.8 kW/m of available wave power annually, and a peak value of 717.2 kW/m in late October. The Gulf deployment location has 1,650 W/m of power available. **Table 4** shows the high level overview of the performance for both locations.

Location	**Wave Power Availability**	CO_2 **Sequestered**	**LCOC**
Juan de Fuca Aquifer	61,800 W/m	0.723 Mt/year	$6.30/t
Gulf of America	1,650 W/m	0.414 Mt/year	$10.21/t

3.1 Power Generation

Using wave power for pumping is an efficient and sustainable method of harnessing the ocean's natural mechanical energy directly, eliminating the need for electrical conversion. Traditional pumping systems often rely on electrical energy, typically generated from fossil fuels or centralized renewable sources, which must then be converted into mechanical energy to function. CASHEW uses the direct motion of waves to drive mechanical components, making the process inherently more efficient. Direct-drive systems are especially advantageous in this context, as they mechanically couple the oscillatory wave motion directly to the pump without intermediate energy conversion steps, reducing energy losses and equipment complexity. This direct coupling enhances reliability and lowers operational costs by minimizing maintenance and eliminating expensive power converters. Economically, wave-powered pumping is particularly suitable for remote or coastal locations where grid access is limited and wave energy is abundant. Compared to solar or wind power, wave energy tends to have higher energy density and greater predictability, providing a steadier and potentially more cost-effective resource [21].

3.2 Sequestration Performance

The expected power efficiency and life expectancy depend on both the wave energy converter (WEC) and the aquifer. Without major losses due to energy conversion, the direct coupling of the CASHEW system allows the majority of the mechanical energy to be used directly to pump the carbon into the aquifer. CASHEW's expected power efficiency can be looked at by seeing power losses throughout the process. The double piston system can provide losses in power due to aspects like friction and compressibility. On the aquifer side, energy losses occur as the pressurized fluid enters the ground, with subsurface resistance and aquifer permeability.

Figure 10, presents the results of the system dynamics model, showing the sequestration rate in kg/s for the two locations over time. If we take a time average of these simulations and convert to tons per year, we see a result of 0.723 Mt/year for the Juan de Fuca location and 0.414 Mt/year for the Gulf location. At these rates, it would take 1383 and 2415 total CASHEW systems globally, respectively. There is opportunity for increased flow rates (therefore reducing the required total units globally) through improved design. Particularly in the RM3 WEC design.

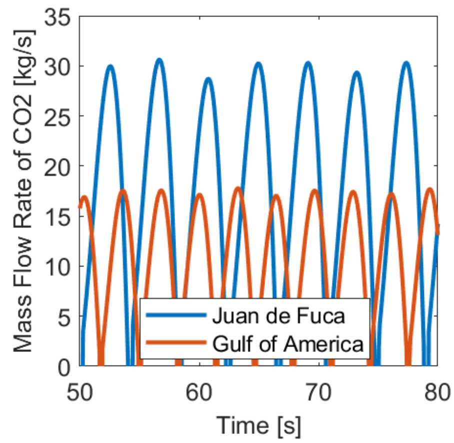

Figure 10: Sequestration rates for the two identified locations.

3.3 Life Expectancy

The projected life expectancy of the WEC is 20 years, with routine maintenance required for moving parts such as pistons and seals. With a CO_2 sequestration rate of 0.723 Mt/year and the Juan de Fuca aquifer having a projected CO_2 storage capacity of 208 Gt it would take approximately 288,000 years before the aquifer is entirely filled. If we were to have a large WEC array in hopes of meeting the goal of 1 Gt/year, the Juan de Fuca aquifer would still take over 200 years to fill. This shows that the limiting factor is the WEC lifetime, meaning the system should be operational for the 20 years before the WEC needs to be replaced.

3.4 Economics

The cost breakdown for the CASHEW system in the two locations is shown in **Table 5**. It is notable that the costs of the system are dominated largely by the WEC costs. Additionally, since a large piece of the WEC costs are due to things such as infrastructure and installation, integrating the system with existing decommissioned off-shore oil rigs (as discussed in the CONCEPT phase) could be beneficial and future work will consist of quantifying these benefits.

Table 5: CASHEW cost breakdown.

Total WEC installation costs ($2025)	$22,763,700
WEC Annual costs ($2025)	$1,576,800
Cost of hydraulic cylinder, RM3 data ($2025)	$35,846.47
Juan de Fuca Pipe ($2025)	$9,984,214.37
Gulf of America Pipe ($2025)	$6,386,550.53
Juan de Fuca Total Annual Cost ($2025)	**$4,552,135.96**
Gulf of America Total Annual Cost ($2025)	**$4,225,624.96**

From these costs and the results from the system dynamics modeling the LCOC for the two locations are $6.30 and $10.21 per ton of CO_2 for the Juan de Fuca and Gulf locations respectively. Notably, both these results meet our requirement of $6 - $12 per ton of CO_2, suggesting that CASHEW presents a competitive option for carbon sequestration.

4. Testing and Validation Planning

In the next phase we plan to make improvements to the model, and then put together a physical proof of concept system. The next steps are presented with a Gantt chart of the timeline along with an estimated budget.

4.1 Next Steps

The next step is to finalize some of the key gaps in the model. For the economic model, a cost estimate for the piston based on design is necessary to fully model design tradeoffs. Some members of the team have built a cost model for a piston based on bore diameter, stroke length, and pressure using ASME BPV code and scaling to match trends found in quotes [22]. Additionally in the

CONCEPT phase, we proposed that integrating the system with repurposed offshore oil rigs could be economically beneficial. The current cost model does not have the ability to investigate the impact so further work to add this as an option to our cost model would allow us to revisit that idea.

The team is also interested in investigating a long time scale aquifer model similar to the work presented by Qin et al. [23], to model how the aquifer pressure would change over time. This would be helpful for evaluating system performance at different stages throughout CASHEW's lifetime. The model could exist on its own to just predict how much different the aquifer pressure would be at different stages, but the aspect of developing this model the team is more interested in is using it in a multi-fidelity lifetime analysis simulation. Instead of only simulating the system dynamics for a few minutes. The results of an initial system dynamics simulation could feed into the aquifer pressure model which would simulate a few months of aquifer filling and then using the new pressure calculated at the end of those few months run the system dynamics simulation again. This process could then iterate for the full 20 year lifetime of the system, or until the system stops being economically beneficial.

Although for the DEVELOP phase, we did not pursue a physical proof of concept prototype, in the next stage this will be achievable given the larger prize pool. A full scale system is not possible, but a scaled down model would be. Our lab has experience prototyping small scale WECs [24, 25]. To complete the prototype a model of an aquifer will be required. The important aspects of the aquifer to model are the large pressure already present in the aquifer and the resistance of the aquifer pores to flow, both of which would need to scale with the prototype. For the large pressure, this could be achieved through a variety of ways, one simple way is a weighted piston cylinder. The weights on the piston are preferred to a spring as they will ensure the pressure does not change as the cylinder fills. For the aquifer resistance, a membrane with a properly scaled permeability could work well. Our lab also does work with ceramic ultrafiltration membranes, which will aid in the design of this component. Then the final piece of the prototype design will be selecting a fluid that scales with the other properties of the system. A Gantt chart showing the plan for the next year is shown in **Figure 11**.

PHASE ONE: August, September, October · PHASE TWO: November, December, January · PHASE THREE: Februry, March, April · PHASE FOUR: May (weeks 1 2 3 4 in each month)

WBS NUMBER	TASK TITLE	DURATION	PCT OF TASK COMPLETE
1	Improve Model		
1.1.1	Piston cost model	3 weeks	0%
1.1.2.1	Research oil rig cost savings	4 weeks	0%
1.1.2.2	Quantify oil rig impacts	4 weeks	0%
1.2.1	Aquifer storage model	9 weeks	0%
1.2.2	Integrate multifidelity model	4 weeks	0%
2	Design Prototype		
2.1	Prototype architecture	4 weeks	0%
2.2.1	Fluid Scaling	2 weeks	0%
2.2.2	Identify Suitable Working Fluid	1 weeks	0%
2.3	Detail design	4 weeks	0%
2.4	Design model aquifer	4 weeks	0%
2.5	Document design	2 weeks	0%
3	Build and Test		
3.1	Order Parts	2 weeks	0%
3.2	Build	5 weeks	0%
3.3	Test in small scale wave tank + pool	3 weeks	0%
3.4	Adjustments	4 weeks	0%
3.5	Data Collection	1 weeks	0%
3.6	Analaysis	2 weeks	0%
4	Documentation and Submit		
4.1	Write Report	2 weeks	0%
4.2	Internal Review	1 week	0%
4.3	Edits	1 week	0%

Figure 11. Gantt chart for the next year.

Beyond the next year, the team hopes work will still happen on this project. One of the major places where we see opportunities for growth with this project is in improving the design of the system. Particularly in the design of the WEC. The RM3 WEC was chosen here because of the great documentation associated with it and the focus on modeling for the DEVELOP phase. However, we believe there is significant room for improvement in system performance. Designing a WEC specifically designed for CASHEW and finding an optimal piston size would both have major positive impacts on system performance and are worthy of attention once the full model is developed and proof of concept is demonstrated.

4.2 Anticipated Challenges

For the economic modeling gaps, while our team has experience building cost models for hydraulic cylinders, we do not have a good understanding of the cost benefits co-locating with an offshore oil rig would bring. In the CONCEPT phase, we had a conversation with Kent Satterlee, the Executive Director at Gulf Offshore Research Institute. Kent manages a few decommissioned offshore oil and gas (O&G) platforms in the Gulf and we will plan to collaborate with Kent to explore the economic advantages of co-locating with these O&G platforms.

The long term aquifer pressure modeling work was started in the DEVELOP phase, but not included in this submission due to technical roadblocks. A lot of the roadblocks experiences are due to our team's inexperience with geological dynamics. To aid in this effort, our team will look to add a student from Cornell's department of Earth at Atmospheric Science, who is more familiar with the type of modeling work required for this project.

Although our lab has experience prototyping and testing small scale WECs, the next stage will require scaling of fluid properties to properly mimic liquid/supercritical carbon dioxide dynamics at our smaller scale. Identifying a good fluid candidate will likely be a major roadblock and challenge. The selection of what material our hydraulic circuit connecting our WEC piston and aquifer model as well as the selection of the proper permeability and membrane to model the aquifer will be heavily dependent on the fluid choice. Cornell has an experimental fluids course and either recruiting a student who has already taken the course to join our team, having some members of the team take the course, or having a discussion with the Professor could all aid in this endeavor.

4.3 Budget

A rough budget for the next phase is shown in **Table 6**. This budget is an anticipated minimum budget, but more funds would allow us to get more time tank testing, more undergraduate researchers, and/or more design flexibility. At this stage there is also quite a bit of uncertainty in this budget, as aside from the WEC prototype, our team does not yet have much knowledge and experience in this sort of work.

Table 6: Budget for next phase.

Item	**Cost**
WEC prototype	$8,000
WEC driven hydraulic cylinder	$7,000
Hydraulics (valves and piping)	$1,000
Membranes for "aquifer"	$5,000
Hydraulic cylinder for "aquifer"	$8,000
Working fluid, "CO_2"	$1,000
Tank testing at UNH	$5,000
Paid undergraduate researchers	$30,000
Contingency	$10,000
Total	**$75,000**

References

[1] J. Niffenegger, D. Greene, R. Thresher, and M. Lawson, *Mission Analysis for Marine Renewable Energy To Provide Power for Marine Carbon Dioxide Removal*. Golden, CO: National Renewable Energy Laboratory, Sep. 2023.

[2] R. J. Rosenbauer, B. Thomas, J. L. Bischoff, and J. Palandri, "Carbon sequestration via reaction with basaltic rocks: Geochemical modeling and experimental results," *Geochimica et Cosmochimica Acta*, vol. 89, pp. 116–133, 2012, doi: 10.1016/j.gca.2012.04.042.

[3] Ocean Visions, *Marine Carbon Dioxide Removal (mCDR)*. [Online]. Available: https://oceanvisions.org/deep-sea-storage/. [Accessed: May 2025].

[4] H. Norton, P. Gillessen, and C. Crawford, "Techno-economic assessment of supercritical, cold liquid, and dissolved CO_2 injection into sub-seafloor basalt," *Carbon Capture Sci. Technol.*, vol. 13, p. 100236, 2024, doi: 10.1016/j.ccst.2024.100236.

[5] J. S. Levine, J. M. Matter, D. Goldberg, A. Cook, and K. S. Lackner, “Gravitational trapping of carbon dioxide in deep sea sediments: Permeability, buoyancy, and geomechanical analysis,” *Geophys. Res. Lett.*, vol. 34, no. 24, Dec. 2007, doi: 10.1029/2007GL031560.

[6] M. H. Rasool and M. Ahmad, “Reactivity of basaltic minerals for CO_2 sequestration via in situ mineralization: A review,” *Minerals*, vol. 13, no. 9, Art. no. 1154, 2023, doi: 10.3390/min13091154.

[7] NREL, *Marine Energy Atlas*. [Online]. Available: https://www.nrel.gov/gis/marine-energy.html. [Accessed: May. 2025].

[8] A. L. Aretxabaleta, Z. Defne, T. S. Kalra, B. O. Blanton, and N. K. Ganju, *Climatological wave height, wave period and wave power along coastal areas of the East Coast of the United States and Gulf of Mexico: U.S. Geological Survey data release*, unpublished, 2022. [Online]. Available: https://doi.org/10.5066/P9HJ0JIQ.

[9] D. S. Goldberg, T. Takahashi, and A. L. Slagle, “Carbon dioxide sequestration in deep-sea basalt,” *Proc. Natl. Acad. Sci.*, vol. 105, no. 29, pp. 9920–9925, 2008, doi: 10.1073/pnas.0804397105.

[10] A. T. Fisher, “Permeability within basaltic oceanic crust,” *Rev. Geophys.*, vol. 36, no. 2, pp. 143–182, 1998, doi: 10.1029/97RG02916.

[11] Bureau of Ocean Energy Management, *Atlantic OCS Aliquots with Sand Resources*, U.S. Department of the Interior, 2025. [Online]. Available: https://catalog.data.gov/dataset/atlantic-ocs-aliquots-with-sand-resources-1e93f. [Accessed: May, 2025].

[12] Bureau of Ocean Energy Management, *Marine Minerals Information System (MMIS)*, U.S. Department of the Interior. [Online]. Available: https://mmis.boem.gov. [Accessed: May, 2025].

[13] A. Wendt, A. Sheriff, C. Y. Shih, D. Vikara, and T. Grant, "A multi-criteria CCUS screening evaluation of the Gulf of Mexico, USA," Int. J. Greenhouse Gas Control, vol. 118, p. 103688, 2022, doi: 10.1016/j.ijggc.2022.103688.

[14] J. Bauer, C. Disenhof, and E. Cameron, *Gulf Of Mexico Reservoir Sands*, 2016. [Online]. Available: https://edx.netl.doe.gov/dataset/gulf-of-mexico-reservoir-sands-2016. [Accessed: May, 2025].

[15] Bureau of Ocean Energy Management, *Atlas of Gulf of America Gas and Oil Sands Data*, 2024.

[16] N. I. Osborn, S. J. Smith, and C. H. Seger, *Hydrogeology, Distribution, and Volume of Saline Groundwater in the Southern Midcontinent and Adjacent Areas of the United States*, USGS Groundwater Resources Program, 2013-2017. Available: http://dx.doi.org/10.3133/sir20135017.

[17] *WEC-Sim (Wave Energy Converter SIMulator)*, Available: https://wec-sim.github.io/WEC-Sim/master/index.html.

[18] National Renewable Energy Laboratory (NREL), *Expert Elicitation for Wave Energy LCOE Futures*, 2022. [Online]. Available: https://docs.nrel.gov/docs/fy22osti/82375.pdf. [Accessed: May 2025].

[19] Sandia National Laboratories, *Reference Model 3: Oscillating Surge Wave Energy Converter*, Marine and Hydrokinetic Data Repository, 2014. [Online]. Available: https://mhkdr.openei.org/submissions/370. [Accessed: May 2025].

[20] M. A. Kaiser and A. G. Pulsipher, "Offshore pipeline construction cost in the U.S. Gulf of Mexico," *Marine Policy*, vol. 75, pp. 95–103, Nov. 2016. [Online]. Available: https://doi.org/10.1016/j.marpol.2016.10.007ScienceDirect

[21] A. F. de O. Falcão, "Wave energy utilization: A review of the technologies," *Renew. Sustain. Energy Rev.*, vol. 14, no. 3, pp. 899–918, Apr. 2010, doi: 10.1016/j.rser.2009.11.003.

[22] N. DeGoede and M. Haji, *A Multidisciplinary Design Optimization Framework for Wave-Driven Desalination Systems*, accepted for publication at the ASME 2025 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE2025), Anaheim, CA, USA, Aug. 17–20, 2025, Paper No. DETC2025-168312.

[23] J. Qin, Q. Zhong, Y. Tang, Z. Rui, S. Qiu, and H. Chen, "CO_2 storage potential assessment of offshore saline aquifers in China," *Fuel*, vol. 341, p. 127681, Jun. 2023, doi: 10.1016/j.fuel.2023.127681.

[24] Vitale, O., A. Brundan, Y. Mandalam, R. McCabe, Haji, M., "Design, manufacturing, and deployment of heterogeneous wave energy converter arrays," in prep, 2025.

[25] Vitale, O., A. Ahmed, Haji, M., "Experimental campaign and open-source dataset generation for novel wave energy converter farms" in prep, 2025.